For teachers only. Do not give this part to students.

Red water traces in celery: does light change water transport?

Distribution, clarification, feedback and timing

1. Study conditions: experimental group uses IBL plus VL; comparison group has no prior lecture, does not operate VL, and learns from A, B and C alone. Allow 15–20 minutes per group, targeting 18 minutes for both; pilot actual time.
2. Minutes 0–4: give out A and B together. Students read concepts, the leaf diagram and worked examples in A themselves. Core teaching is available from the start.
3. Minutes 4–7: B1–B2 supported practice, specified setups and fair comparisons. Students may consult A; no design from scratch.
4. Minutes 7–15: B3–B5 ruler reading, calculation, plotting and analysis. Save first B answers before giving C; record unfinished work within the common time limit, not automatic zero.
5. Minutes 15–18: check with C, explain using data and correct errors. A remains available. D and the full pack are teacher-only; do not distribute C or D at the start.
6. Teachers clarify procedures such as page order, circling, units or ruler position only. No prior lecture or answer hints before B is completed. Record clarifications and actual time; use the standard C key for feedback.
7. Both groups share concepts, skill goals and model data with similar time; teaching order and interaction differ. Hypothesis evaluation after teaching is not original hypothesis performance. C provides the full key; D supplies alignment and feedback.
8. Use separate common tests such as proposal SPSAI / CKT for effectiveness, not activity totals. Researchers must confirm test versions, language, raters, feasibility within 15–20 minutes, and grayscale printing.

Paper ruler reading is not full experimental operation. Do not directly compare paper scores with VL original prediction, operation or completion scores; evaluate effectiveness through separate common tests.

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Feedback by area (not a group-effectiveness total)

Record areas separately, without a total compared with the VL 32-point score. Missing responses are unfinished / awaiting assessment. Paper reading is not full practical operation. Use separate common tests for effectiveness.

Area	2: sufficient	1: partial	0: needs correction
Observation	Describes change and separates inference	Some visible change	Observation needs correction
Classification	All seven factors correct	Some correct	Classification needs correction
Understanding inquiry	Identifies testable (可測試的) factor, outcome and fair comparison reasons	Incomplete reasons	Confuses factors / comparisons
Reading and recording	All 16 readings and units correct	Some readings correct	Reading method needs correction
Inference	Both comparisons use data and limits	Correct trend; incomplete evidence	Does not match data
Communication	Four speeds, both axes and four points correct; D separate	Some calculations / points correct	Method needs correction
Concepts	Links light, transpiration pull (蒸騰拉力) and xylem (木質部) transport; distinguishes processes	Some mechanism correct	Confuses transport and transpiration (蒸騰)
Checking and correction	Identifies and reasonably corrects original error	Incomplete reason	Copies key only / keeps error

Using the complete answer key

C1 contains observation, hypothesis evaluation, variables, assumptions, setups, all readings, speeds and choice answers. C2 contains the graph, concept explanation and correction examples. Diagnose original B responses before reviewing C corrections; copying answers is not understanding.

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Question alignment • 1–5

VL original	Treatment	Paper task	Shared goal	Reason
Your first observations	Retain	B1 Describe visible changes in the trace; keep explanations separate from observations.	Observation	Separate observation and inference
If light intensity (光強度) increases, the upward speed of the red water trace will (increase / decrease / show no clear change).	Adapt	A2 → B1 Evaluate “Celery with a larger leaf area is better.” Can this statement be tested directly as written? (yes / no) Identify the factor and replace “better” with a measurable result. This is evaluation after learning, not an original prediction.	Testable (可測試的) hypotheses	Teach, then evaluate; not original prediction
My reason	Adapt	B1 Evaluate “Celery with a larger leaf area is better.” Can this statement be tested directly as written? (yes / no) Identify the factor and replace “better” with a measurable result. This is evaluation after learning, not an original prediction.	Explain testability	State factor and measured outcome
Identify the variables	Retain classification	A2 → B1 For the light comparison using A, B and C, tick one column for each factor.	Classify variables	Same seven factors and three classes
What assumptions does this investigation make?	Retain	B2 What assumptions does this investigation make? Changing lamp distance may change temperature. Why monitor and keep all setups at 25°C?	Assumptions and fairness	Keep choices and explain temperature control

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Question alignment • 6–10

VL original	Treatment	Paper task	Shared goal	Reason
How would you plan a comparison setup to find out whether leaves help with water transport?	Equivalent adaptation	B2 For light intensity (光強度) use (A/B/C / A/D); for leaf presence use (B/D / A/D). Why is A/D not a fair comparison of leaf presence?	Fair comparisons and controls	Identify comparisons in specified setups
Include the celery, leaves, red water, lamps, distances and measurement positions. Put 2 cm of each celery leaf stalk into the red water. Setup labels / written design	Complete labels	B2 ① points to the long stalk, ② to the liquid surface, ③ to the immersed part. Complete: plant part and water-carrying tissue for ①; ruler reading at ②; immersion depth for ③.	Understand setups and conditions	Read and justify controls instead of free design
All four setups run together. Take measurements every 10 minutes of simulated time.	Static	B2–B4 Read 0 min in B2 and 10, 20 and 30 min in B3–B4; fill the B4 table. You may use methods in A; do not read C yet.	Identify ruler procedure	Not full practical operation
Your measurement record (0/10/20/30 min)	Static ruler reading	A2 → B2–B4 Read 0 min in B2 and 10, 20 and 30 min in B3–B4; fill the B4 table. You may use methods in A; do not read C yet.	Read and record	Same four times; neutral example first
Calculate the average upward speed	Retain	A2 → B4 Average upward speed = (height at 30 min − height at 0 min) ÷ 30. Use two decimal places and cm/min.	Calculate mean speed	Same formula and two decimals

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Question alignment • 11–13

VL original	Treatment	Paper task	Shared goal	Reason
Light intensity (光強度) and the upward speed of the red water trace	Paper plotting	A2 → B5 Plot four points using your calculations. X: light intensity (光強度) (lux); Y: average upward speed (cm/min). Join A, B and C in light-intensity order to show a reasonable trend. Mark D separately; do not join it into that trend.	Graph communication	Same axes; D separate
As light intensity (光強度) increases, the average upward speed of the red water trace usually will (increase / decrease / show no clear change / show no consistent trend).	Retain with evidence	B5 As light intensity (光強度) increases, the average upward speed of the red water trace usually will (increase / decrease / show no clear change / show no consistent trend). Use data from A, B, C and B/D to support the conclusions.	Light trend	Same choices and data
At the same light distance of 20 cm, the average upward speed of the water trace in Setup B (with leaves), compared with Setup D (without leaves), is (faster / slower / similar). This comparison supports the idea that leaves (help water move upwards / stop water moving upwards / have no effect on water moving upwards).	Retain	B5 At the same light distance of 20 cm, the average upward speed of the water trace in Setup B (with leaves), compared with Setup D (without leaves), is (faster / slower / similar). This comparison supports the idea that leaves (help water move upwards / stop water moving upwards / have no effect on water moving upwards). Use data from A, B, C and B/D to support the conclusions.	B/D comparison and leaf role	Includes original leafConclusion item

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Question alignment • 14–15

VL original	Treatment	Paper task	Shared goal	Reason
We measured only the height of the red water trace. We did not measure water vapour (水汽) lost from the leaves. Which statement is most suitable?(The red water trace gives an indirect clue about water uptake and upward water transport. Its speed is not the rate of water loss from leaves, or the transpiration rate (蒸騰速率). / The upward speed of the water trace is the rate of water vapour (水汽) loss from leaves and can be used directly as the transpiration rate (蒸騰速率). / If the setups with leaves are faster than the setup without leaves, this fully proves that water vapour (水汽) is lost only through stomata (氣孔).)	Retain with limits	B5 → C1 We measured only the height of the red water trace. We did not measure water vapour (水汽) lost from the leaves. Which statement is most suitable?(The red water trace gives an indirect clue about water uptake and upward water transport. Its speed is not the rate of water loss from leaves, or the transpiration rate (蒸騰速率). / The upward speed of the water trace is the rate of water vapour (水汽) loss from leaves and can be used directly as the transpiration rate (蒸騰速率). / If the setups with leaves are faster than the setup without leaves, this fully proves that water vapour (水汽) is lost only through stomata (氣孔).) Can three light-intensity points give exact speeds for all untested values? (yes / no)	Evidence limits	Transport is not transpiration (蒸騰); no exact extrapolation
Do your data support your original hypothesis? Use ideas from the key learning points above to improve your original explanation. Explain how it relates to your data.	Adapt	C2 Use A/B/C and B/D data to explain light, transpiration (蒸騰) and xylem (木質部) transport, including suitable conditions and evidence limits. Choose a wrong or incomplete answer of your own: state it and correct it with data or concepts. If none, correct “trace speed is transpiration rate (蒸騰速率)”.	Explain and correct	Use data and correct mistakes

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